Midterm

Graded

Student

HARSH DADHICH

Total Points

90 / 100 pts

Question 1

Multiple Choice (Single Answer) **40** / 40 pts

1.1 a **10** / 10 pts

✔ **– 0 pts** Correct

– 10 pts Click here to replace this description.

1.2 b **10** / 10 pts

✔ **– 0 pts** Correct

– 10 pts incorrect

1.3 c **10** / 10 pts

✔ **– 0 pts** Correct

– 10 pts incorrect

1.4 d **10** / 10 pts

✔ **– 0 pts** Correct

– 10 pts incorrect

Question 2

Bigram Language Model **30** / 30 pts

2.1 a **10** / 10 pts

✔ **– 0 pts** Correct

– 5 pts major mistake (only one answer right)

– 10 pts Incorrect

2.2 b **10** / 10 pts

✔ **– 0 pts** Correct

– 5 pts major mistake

– 10 pts Incorrect

2.3 c **10** / 10 pts

✔ **– 0 pts** Correct

– 3 pts minor mistake

– 6 pts major mistake

– 10 pts Incorrect

Question 3

Decoding **20** / 20 pts

3.1 a **10** / 10 pts

✔ **– 0 pts** Correct

– 3 pts minor mistake

– 6 pts major mistake

– 10 pts Incorrect

3.2 b **10** / 10 pts

✔ **– 0 pts** Correct

– 3 pts minor mistake

– 6 pts major mistake

– 10 pts Incorrect

Question 4

Transformer

0 / 10 pts

- **− 0 pts** Correct
- **− 3 pts** minor mistake
- **− 6 pts** major mistake
- ✔ **− 10 pts** Incorrect

CS 263: Natural Language Processing

Midterm Exam Answer Sheet   Spring 2026

Name: Harsh Dadhich   **Student ID:** 505960902

Circle the correct choice for each multiple-choice question. For numerical answers, round to two decimal places or provide an exact fraction. Show your work for partial credit.

Note: Only the answer sheet will be graded. Please show all your work here.

Section 1: Multiple Choice (Single Answer) [40 pts]

(a) [8 pts]	**A**	**B**	**C**	**D**
(b) [8 pts]	**A**	**B**	**C**	**D**
(c) [8 pts]	**A**	**B**	**C**	**D**
(d) [8 pts]	**A**	**B**	**C**	**D**

Section 2: Bigram Language Model [30 pts]

(a) [10 pts]

$$\frac{c(<s> < I)}{c(<s>)}$$

$$\frac{c(I < am)}{c(am)} = \frac{3}{3}$$

$P(\mathrm{I} \mid \langle \mathrm{S} \rangle) =$ $\frac{2}{3}$ $P(\mathrm{am} \mid \mathrm{I}) =$ 1

(b) [10 pts]

$$\frac{2}{3} \cdot 1 \cdot \frac{1}{3} \cdot \frac{1}{2} = \frac{2}{18} = \frac{1}{9}$$

$P(\langle \mathrm{S} \rangle \text{ I am Sam } \langle /\mathrm{S} \rangle) =$ $\frac{1}{9}$

(c) [10 pts]

$$\frac{2+1}{3+6} \cdot \frac{3+1}{3+6} \cdot \frac{1+1}{3+6} \cdot \frac{1+1}{2+6} = \frac{3}{9} \cdot \frac{4}{9} \cdot \frac{2}{9} \cdot \frac{2}{8} = X$$

$P_{\text{add-1}}(\text{legend} \mid \text{am}) =$ $\frac{2}{9}$

$$\frac{1+1}{3+6} = \frac{2}{9}$$

Section 3: Decoding [20 pts]

(a) [10 pts] Greedy search sequence:

Using argmax per step ; fast , no undo , globally suputimal

<start> → She (-1) → ran (-1) → into (-5) → a

Ans: She ran into a

(b) [10 pts] Beam search ($k = 2$) sequence:

top-2 partial hypothese,

step 1 → She & he → step 2 → she ran, he ran, → step 3

she ran, into

he ~~ran~~ down ↗
fell

step 4

he fell

down

stairs

Ans: He fell down stairs

Section 4: Transformer [10 pts]

Compute O:

output matrix

$$\text{Attn} = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V$$

need to compute for QK^T

$$\begin{bmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \end{bmatrix} = u \cdot \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 2 & 2 \end{bmatrix}$$

$QK^T = (L \cdot D)(D \cdot L) = L \cdot L$

$O(L^2)$

$$O = \begin{pmatrix} \_\_\_\_ & \_\_\_\_ \\ \_\_\_\_ & \_\_\_\_ \\ \_\_\_\_ & \_\_\_\_ \end{pmatrix}$$

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